MATHEMATICAL GAME THEORY

Midterm Examination: Solutions and Rubric

Fall 2024 · Prof. L. Nakamura

Grading philosophy. For proofs, the rubric lists the essential steps. A proof that is correct but organized differently receives full credit. A proof that asserts a key step without justification (e.g. "clearly the mixture is better") loses the points attached to that step. Do not deduct for notation that differs from the solutions if it is defined.

Problem 1 *Dominated strategies and best responses* (12 points)

(a) *Step 1: dominance extends to mixed opponents.* For any $\sigma_{-i} \in \prod_{j \neq i} \Delta(S_j)$,

$$u_i(\tau_i, \sigma_{-i}) - u_i(s_i, \sigma_{-i}) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) \big[u_i(\tau_i, s_{-i}) - u_i(s_i, s_{-i}) \big] > 0,$$

since each bracket is positive and the weights are nonnegative and sum to 1.

Step 2. Let σ^* be a Nash equilibrium and suppose, for contradiction, that $\alpha := \sigma_i^*(s_i) > 0$. Define

$$\sigma_i' = \sigma_i^* - \alpha\, \delta_{s_i} + \alpha\, \tau_i,$$

where δ_{s_i} is the point mass on s_i. Every coordinate of σ_i' is nonnegative (the s_i coordinate equals $\alpha\tau_i(s_i) \geq 0$) and the coordinates sum to 1, so $\sigma_i' \in \Delta(S_i)$. Since u_i is linear in player i's own mixed strategy,

$$u_i(\sigma_i', \sigma_{-i}^*) - u_i(\sigma_i^*, \sigma_{-i}^*) = \alpha \big[u_i(\tau_i, \sigma_{-i}^*) - u_i(s_i, \sigma_{-i}^*) \big] > 0$$

by Step 1. This contradicts σ_i^* being a best response to σ_{-i}^*. Hence $\sigma_i^*(s_i) = 0$. □

RUBRIC. 2 pts: Step 1 (extension from pure to mixed opponent profiles by linearity). 2 pts: construction of σ_i' and verification that it is a mixed strategy. 2 pts: payoff comparison and contradiction. A proof that only treats the case where τ_i is a pure strategy: at most 4/6.

(b) Row chooses U or D, Column chooses L or R:

	L	R
U	1, 1	1, 1
D	1, 1	0, 1

D is weakly dominated by U. In (D, L), Row earns 1 from either row and Column earns 1 from either column, so (D, L) is a Nash equilibrium in which the weakly dominated strategy D is played with probability 1.

RUBRIC. 1 pt: valid weak (not strict) dominance. 2 pts: exhibits and verifies an equilibrium using the dominated strategy.

(c) Against L, U earns $3 > 2$; against R, D earns $3 > 2$; so M is a best response to neither pure strategy. Against q: $u(U) = 3q$, $u(M) = 2$, $u(D) = 3(1-q)$. M is a best response iff $2 \geq 3q$ and $2 \geq 3 - 3q$, i.e. $\boxed{q \in [\tfrac{1}{3}, \tfrac{2}{3}]}$. M is **not** strictly dominated: if some τ strictly dominated M, then by Step 1 of (a), $u(\tau, q = \tfrac{1}{2}) > u(M, q = \tfrac{1}{2}) = 2$; but against $q = \tfrac{1}{2}$ the payoffs are $\tfrac{3}{2}, 2, \tfrac{3}{2}$, so no mixed strategy earns more than 2.

RUBRIC. 1 pt: not a best response to pure strategies. 1 pt: interval $[\frac{1}{3}, \frac{2}{3}]$. 1 pt: not strictly dominated, with a proof (checking only pure U and D is insufficient).

Problem 2 *Zero-sum games* (14 points)

(a) For all $x' \in \Delta_m$ and $y' \in \Delta_n$,

$$\min_y x'^\top A y \;\le\; x'^\top A y' \;\le\; \max_x x^\top A y'.$$

(The extrema exist because $x^\top A y$ is continuous and the simplices are compact.) The left side does not depend on y' and the right side does not depend on x'. Taking the maximum over x' on the left and then the minimum over y' on the right yields the claim. □

RUBRIC. 2 pts: the chain of inequalities for fixed x', y'. 2 pts: correct order of taking max and min. Invoking the minimax theorem earns 0.

(b) Let $v = (\sum_i 1/a_i)^{-1} > 0$ and $x_i = y_i = v/a_i$. These are nonnegative and $\sum_i v/a_i = 1$, so x, y are mixed strategies.

- Against pure column j, $x^\top A e_j = a_j x_j = v$. Hence $x^\top A y' = \sum_j y'_j v = v$ for every $y' \in \Delta_n$: Row guarantees v, so $\max_x \min_y x^\top A y \ge v$.
- Against pure row i, $e_i^\top A y = a_i y_i = v$. Hence $x'^\top A y = \sum_i x'_i v = v$ for every $x' \in \Delta_n$: Column holds Row to v, so $\min_y \max_x x^\top A y \le v$.

Combining with (a): $v \le \max\min \le \min\max \le v$, so all are equal to v, which is therefore the value, and x, y are optimal. □

RUBRIC. 1 pt: x, y are valid distributions. 2 pts: Row's guarantee. 2 pts: Column's guarantee. 1 pt: conclusion via (a) (or the minimax theorem).

(c) Let y be any optimal strategy for Column. Then $e_i^\top A y = a_i y_i \le v$ for every i, so $y_i \le v/a_i$. Summing,

$$1 = \sum_i y_i \le v \sum_i \frac{1}{a_i} = 1,$$

so every inequality holds with equality and $y_i = v/a_i$ for all i: the optimal strategy is unique. (The same argument with $a_j x_j \ge v$ shows Row's optimal strategy is unique.)

For $(2,3,6)$: $\sum 1/a_i = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$, so $\boxed{v = 1}$ and $x = y = \boxed{(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})}$.

RUBRIC. 3 pts: uniqueness proof (the summing argument). 1 pt: numerical answer.

Problem 3 *Symmetric* 2×2 *games* (12 points)

(a) For $p \in [0,1]$ write $u(s,p) = p\,u(s,A) + (1-p)\,u(s,B)$ and let

$$D(p) = u(A,p) - u(B,p),$$

the advantage of A over B against an opponent who plays A with probability p. D is affine in p.

- If $D(1) \geq 0$: A is a best response to A, so (A,A), i.e. $p^* = 1$, is a symmetric equilibrium.
- If $D(0) \leq 0$: B is a best response to B, so (B,B), i.e. $p^* = 0$, is a symmetric equilibrium.
- Otherwise $D(1) < 0 < D(0)$. Since D is affine, $p^* = \dfrac{D(0)}{D(0) - D(1)} \in (0,1)$ satisfies $D(p^*) = 0$. Against p^* both pure strategies earn the same payoff, so every mixed strategy (in particular p^*) is a best response, and (p^*, p^*) is a symmetric equilibrium.

The three cases are exhaustive. □

RUBRIC. 2 pts: definition of D and its affinity/continuity. 1 pt each: the two pure cases. 2 pts: interior case (existence of the root and why p^* is a best response to itself). 1 pt: exhaustiveness. Invoking Nash's theorem plus a symmetry argument without proof: at most 2/7.

(b) $D(p) = \big[p + 5(1-p)\big] - \big[3p + 2(1-p)\big] = 3 - 5p$.

- Pure: the best response to A is B ($3 > 1$) and to B is A ($5 > 2$). So (A,B) and (B,A) are equilibria; (A,A) and (B,B) are not.
- If one player uses a pure strategy, the other has a unique best response, so an equilibrium with exactly one mixing player does not exist.
- If both mix, each must be indifferent, so the opponent's probability solves $D(p) = 0$: $p = \frac{3}{5}$ for both.

Nash equilibria: $\boxed{(A,B),\ (B,A),\ (\tfrac{3}{5}, \tfrac{3}{5})}$; in the mixed one each player earns $u(A, \frac{3}{5}) = \frac{3}{5} + 2 = \frac{13}{5}$.

RUBRIC. 2 pts: two pure equilibria. 2 pts: mixed equilibrium $p = \frac{3}{5}$. 1 pt: argument that there are no others.

Problem 4 *Potential games* (12 points)

(a) S is finite, so Φ attains its maximum at some s^*. For any i and s_i', $u_i(s_i', s_{-i}^*) - u_i(s^*) = \Phi(s_i', s_{-i}^*) - \Phi(s^*) \leq 0$. So no player can gain by deviating: s^* is a pure Nash equilibrium. □

RUBRIC. 1 pt: existence of a maximizer. 2 pts: the deviation inequality.

(b) Along a strict improvement path $s^0, s^1, \ldots$, each step satisfies $\Phi(s^{t+1}) - \Phi(s^t) = u_i(s^{t+1}) - u_i(s^t) > 0$ for the moving player i. So Φ strictly increases, no profile repeats, and the path has at most $|S| - 1$ steps. □

RUBRIC. 1 pt: Φ strictly increases. 1 pt: finiteness from $|S| < \infty$.

(c) *Potential.* Suppose a commuter moves from A to B, so $(n_A, n_B) \to (n_A - 1, n_B + 1)$. Her cost changes by $c_B(n_B + 1) - c_A(n_A)$. In Ψ, the A-sum loses its last term $c_A(n_A)$ and the B-sum gains the term $c_B(n_B + 1)$, so Ψ changes by exactly the same amount. A move from B to A is symmetric. Thus $-\Psi$ is an exact potential for the payoffs $-c$.

Values.

n_A	0	1	2	3
Ψ	$4+5+6=15$	$2+(4+5)=11$	$(2+4)+4=10$	$2+4+6=12$
total cost	18	12	12	18

(total cost $= n_A\, c_A(n_A) + n_B\, c_B(n_B) = 2n_A^2 + n_B(n_B + 3)$)

Equilibria. A unilateral switch changes n_A by ± 1 and changes the switcher's cost by the change in Ψ. So a profile is an equilibrium iff $\Psi(n_A) \le \Psi(n_A \pm 1)$, which holds only for $n_A = 2$. The pure Nash equilibria are the $\boxed{3}$ profiles with exactly two commuters on A and one on B (each pays 4). Direct check: an A-user who switches pays $c_B(2) = 5 > 4$; the B-user who switches pays $c_A(3) = 6 > 4$.

Efficiency. Equilibrium total cost is 12, which equals the minimum total cost; the price of anarchy is 1 here. (The profile with $n_A = 1$ also costs 12 in total but is not an equilibrium: a B-user pays 5 and would switch to A to pay 4.)

RUBRIC. 2 pts: potential identity. 1 pt: Ψ values. 2 pts: all equilibria (the three profiles, or "$n_A = 2$" with the count). 2 pts: total-cost comparison.

Problem 5 *A divisor game* (10 points)

(a) Every move removes at least the chosen element, so $|S|$ strictly decreases and at most $|S_0|$ moves are possible. Since 1 is a multiple only of itself, $1 \in S$ until someone chooses it; so the game ends exactly when 1 is chosen, and it has a winner and a loser (no draws). It is a finite two-player game of perfect information without draws, so by Zermelo's theorem exactly one player has a winning strategy.

RUBRIC. 1 pt: finiteness. 1 pt: no draws plus Zermelo.

(b) The continuation of the game depends only on the current set S and on who is to move. Call S a *P-position* if the player to move from S loses under optimal play.

Suppose, for contradiction, that Player 2 has a winning strategy. Let Player 1 open with N; since N is the only multiple of N in S_0, the position becomes $S_0 \setminus \{N\}$. Player 2's winning strategy answers with some d. If $d = 1$, Player 2 loses at once, which is impossible for a winning strategy; so $d \neq 1$. The resulting position is

$$T = S_0 \setminus \big(\{N\} \cup \{e \in S_0 : d \mid e\}\big) = S_0 \setminus \{e \in S_0 : d \mid e\},$$

because $d \mid N$. Player 2 wins from here with Player 1 to move, so T is a P-position. But Player 1 could have opened with d, reaching the same position T with Player 2 to move; then Player 1

wins by following Player 2's winning continuation from T. So Player 1 has a winning strategy, contradicting the assumption. By (a), Player 1 has a winning strategy. □

RUBRIC. 1 pt: sets up the contradiction. 1 pt: effect of opening with N. 2 pts: key observation that N is a multiple of d, so the positions coincide. 1 pt: handles $d = 1$ and concludes.

(c) $N = p^k$: $S_0 = \{1, p, \ldots, p^k\}$. Player 1 chooses p, which removes every element except 1; Player 2 must choose 1 and loses.

$N = pq$: Player 1 chooses pq, leaving $\{1, p, q\}$. Player 2 must choose 1 (and lose), or p, or q; say p (the case q is symmetric), leaving $\{1, q\}$. Player 1 chooses q, leaving $\{1\}$, and Player 2 must choose 1 and loses.

RUBRIC. 1 pt: prime-power case. 2 pts: $N = pq$ case with Player 2's possible replies handled.